Valuation & Corporate Finance

A Plain-English Study Guide

From class notes → structured understanding

What this guide covers

- **Part 1 —** Unlevering & relevering beta: where the CAPM risk formulas come from
- **Part 2 —** Building Free Cash Flow to the Firm (FCFF) from the balance sheet
- **Part 3 —** Cross-border capital budgeting: tax, currency, Fisher effect & country risk

Every formula in your notes is derived, not just stated. Read top to bottom — each part builds on the one before.

Contents

How the Three Parts Fit Together

Before any detail, hold this map in your head — it is the forest that all the trees below belong to. Every fragment in your notes lives somewhere on it.

> **The one-paragraph map**
>
> **Beta (Part 1)** measures the *risk* of a business, which sets its **cost of capital** (the discount rate). **FCFF (Part 2)** measures the *cash flows*. Discount the cash flows at the cost of capital and you get the **value of the operating business**. Add back **excess cash and other non-operating assets** and you get **firm value**. When the cash flows are in a **foreign currency (Part 3)**, you bridge currencies using the Fisher effect and adjust the discount rate for country risk — and prove that the currency you chose to work in does not change the answer.

You were never missing understanding. You were missing this connective tissue. Let us build it, one part at a time.

PART 1

Unlevering and Relevering Beta

The one idea that makes everything click

A company's **operating assets** have their own inherent riskiness — how much the business's cash flows swing with the overall economy. Call that the **asset risk**, measured by **asset beta (β_a)**, also called **unlevered beta**.

Now pile **debt** on top of that business, and the equity holders' risk goes *up*. Why? Because debt holders get paid first — interest is a fixed promise. Whatever is left over, the volatile residual, belongs to equity. So leverage **concentrates** the business's risk onto a smaller equity base. More debt → riskier equity → higher **equity beta (β_e)**.

> **That is the entire intuition.** Everything else on this page is just bookkeeping around this one sentence: leverage does not create risk, it re-distributes the business's risk onto the equity holders.

Where your five formulas come from

Your notes list $\beta_e = \beta_a(1 + D/E)$ and $\beta_a = w_d \cdot \beta_d + w_e \cdot \beta_e$. Here is the derivation, and it is simpler than it looks. Start from the fact above: the assets are funded by debt + equity. So the risk of the assets must be the **weighted average** of the risk of the debt and the risk of the equity that funds them:

$$\beta(\text{asset}) = [D \,/\, (D+E)] \times \beta(\text{debt}) \;+\; [E \,/\, (D+E)] \times \beta(\text{equity})$$

Your formula #5 — the risk of what the company owns = the blended risk of who paid for it

You cannot create or destroy risk by relabeling who holds the claim — so the asset's risk has to equal the weighted blend of the claim-holders' risks.

The CAPM assumption: why β(debt) = 0

You asked: *in CAPM we assume β(debt) = 0, why?* Debt beta measures how much the *debt's* return swings with the market. For an investment-grade company, debt is nearly risk-free — lenders get their fixed interest whether the economy booms or busts. Its return barely moves with the market, so **β(debt) ≈ 0**. We set it to zero as a simplifying assumption.

Now watch what happens to the formula when β(debt) = 0 — the debt term simply vanishes:

$$\beta(\text{asset}) = [E \,/\, (D+E)] \times \beta(\text{equity})$$

Rearrange to solve for equity beta:

$$\beta(\text{equity}) = \beta(\text{asset}) \times (D+E)/E = \beta(\text{asset}) \times (1 + D/E)$$

Your formula #2 — it fell straight out once we killed the debt term

Add taxes — interest is tax-deductible, which softens the leverage effect — and you get the full textbook version used in practice:

$$\beta(\text{equity}) = \beta(\text{asset}) \times [\, 1 + (1 - t) \times D/E \,]$$

The relevering formula (t = tax rate)

Unlevering vs relevering — what you are actually doing

This is the practical move, and the reason the formulas exist at all. You want the beta of your **target company**, but it is private — no stock price, so no beta to measure. So you **borrow** risk from listed peers, in three steps:

1. **Unlever:** take each comparable's *equity* beta (measurable from its stock) and strip OUT its leverage to get its *asset* beta — the pure business risk. $\beta_a = \beta_e \div (1 + (1-t)D/E)$
2. **Average:** average those asset betas across peers (your formulas #3 and #4). This gives the industry's pure business risk, cleaned of each peer's individual debt load.
3. **Relever:** take that clean industry asset beta and add BACK *your target's* leverage. $\beta_e = \beta_a \times (1 + (1-t)D/E)$

The altitude analogy

Think of it like adjusting race times for altitude. Every peer reports its running time at a different altitude (leverage). You strip each back to sea-level pace (asset beta), average the sea-level paces, then re-adjust for *your* mountain's altitude. **Unlever = go down to sea level. Relever = climb your specific mountain.**

Two loose ends from your notes

"In D/E we take net debt" — yes, and why

Net debt = total debt − cash. We use net debt because cash is a *negative* borrowing. A company with ₹100 of debt and ₹100 of cash could repay the debt tomorrow, so its *effective* leverage is zero. Cash offsets debt. (This connects directly to Part 2, where cash gets special handling.)

"How is valuation conservative if we use CAPM?"

CAPM tends to produce a **relatively high cost of equity**, because it fully prices market risk through beta and the equity premium. A higher discount rate means you divide future cash flows by a bigger number → **lower present value**. A lower valuation is a *conservative* valuation — you are less likely to overpay. That is the sense in which CAPM is conservative: it errs toward caution.

PART 2

Building FCFF from the Balance Sheet

This is the heart of valuation. Your notes have every piece — the task is to put them in order. FCFF stands for **Free Cash Flow to the Firm**: the cash the operating business generates that is available to *all* investors (debt and equity), before financing.

Step 1 — What drives value?

Your notes open with cash flow, cost of capital, growth rate, competitive advantage. That is exactly right — these are the **four value drivers**. A company is worth more if it (a) throws off more cash, (b) is less risky (lower cost of capital), (c) grows faster, and (d) can *sustain* its advantage so competitors do not erode those cash flows. Everything in valuation reduces to these four.

Step 2 — Firm cash flow vs equity cash flow

Your note: *cash flow for assets → start from EBIT; cash flow for equity → start from PAT.* Exactly right, and here is why.

Starting point	Represents	Leads to	Whose cash
EBIT	Profit the operating assets earn, before interest & tax	**FCFF**	Everyone (debt + equity)
PAT	Profit left after interest is paid to lenders	**FCFE**	Shareholders only

EBIT is before anyone is paid, so it is the natural start for firm-level cash. PAT is *after* interest, so it belongs to equity. **Where you start signals whose cash you are computing.**

Step 3 — Accrual vs cash (your Revenue = 1000 example)

Accounting profit is on an **accrual basis** — it records a sale when it is *made*, not when the *cash arrives*. But valuation runs on **cash**, because you cannot spend accounting profit. So we must convert accrual profit into cash. Your worked numbers:

Item	Value	Meaning
Revenue	1,000	Sales booked (accrual)
EBIT	500	Accrual operating profit
Opening A/R	200	Owed by customers at start
Closing A/R	300	Owed by customers at end
Profit	**500**	On paper
Cash	**400**	Profit − ΔA/R = 500 − 100

A/R is money customers owe but have *not paid yet*. It rose 200 → 300, meaning **100 of this year's sales were on credit — booked as profit but no cash came in.** So Cash = Profit − ΔA/R = 500 − 100 = **400**. The rise in receivables is a **use of cash**. This is exactly why we adjust for changes in **Net Working Capital** — receivables, inventory, payables. Growing a business *ties up* cash, and that must be subtracted.

Step 4 — FCFF takes only TWO cash flows

Your note: *free cash flow only takes operating and investing*. Correct — and notice what is **excluded: financing** (debt raised or repaid, interest, dividends). That exclusion is deliberate: it is what makes FCFF "unlevered" — blind to how the company is financed. This ties straight back to Part 1's asset-vs-equity split.

The **investing** adjustments are your two lines:

- **+ Depreciation** — it was subtracted to get EBIT, but it is *non-cash* (no money actually left), so we add it back.
- **− CapEx** — actual cash spent on new equipment and assets; real money out.

Step 5 — Your gross-book-value insight (this is sharp)

Your note observed: *if gross book value rises YoY, CapEx increased (assets added); if accumulated depreciation rises YoY, depreciation increased; so net fixed assets changes as per CapEx and depreciation.* This is precisely how you read CapEx and depreciation off two balance sheets when the cash-flow statement is not handed to you:

$$\text{Net Fixed Assets} = \text{Gross Book Value} - \text{Accumulated Depreciation}$$

- Gross book value **up** → the firm *bought* assets → that is **CapEx**.
- Accumulated depreciation **up** → assets got worn down → that is **Depreciation**.
- So the change in Net Fixed Assets = **CapEx − Depreciation**. You understood the mechanical link perfectly.

Step 6 — Rearranging the balance sheet (the key manoeuvre)

Your note: *move everything from the liability & equity side to the asset side except long-term debt and equity; CA − CL makes NWC; make NFA; make NOA.* Here is the picture. A normal balance sheet is **Assets = Liabilities + Equity**. We reorganize it into **invested capital** — what the long-term investors actually funded.

Starting point: Assets = Current Liabilities + Long-term Debt + Equity. Move the current (operating) liabilities over and net them against current assets:

$$\text{NWC} + \text{NFA} + \text{NOA} = \text{Long-term Debt} + \text{Equity}$$

left side = operating invested capital; right side = the long-term capital that funded it

That is why we move everything *except* long-term debt and equity: we are isolating what investors financed. (NWC = Net Working Capital = CA − CL; NFA = Net Fixed Assets; NOA = other Net Operating Assets such as operating intangibles.)

Step 7 — Why all THREE (ΔNWC, ΔNFA, ΔNOA) are deducted

You asked directly why all three get subtracted. Here is the clean answer. Start from **NOPAT** (EBIT after tax — the operating profit). To turn it into cash, subtract every rupee the business had to **reinvest** to keep running and growing. Each of the three is a form of reinvestment:

- **− Change in NWC** — growing sales tie up more cash in receivables and inventory (your A/R example). Cash goes *in* → subtract.
- **− Change in NFA (net CapEx)** — the business must buy and replace equipment. Cash goes *in* → subtract.
- **− Change in NOA** — any *other* operating assets that grow also absorb cash → subtract.

The unifying principle

Any increase in operating invested capital is cash the business consumed rather than handed to investors. All three are just different buckets of "cash ploughed back in." That is why all three are deducted.

$$\text{FCFF} = \text{NOPAT} - \Delta\text{NWC} - \Delta\text{NFA} - \Delta\text{NOA}$$

This is the *same* FCFF as the income-statement version EBIT(1−t) + Depreciation − CapEx − ΔNWC — two routes, one answer, because ΔNFA = CapEx − Depreciation.

Step 8 — Cash: the tricky one

This is the subtlest part of your notes and genuinely hard, so slowly. Cash comes in **two flavours**:

- **Operating cash** — the minimum the business needs to function day-to-day (float in tills, payroll buffer). This is part of working capital — it is *working*.
- **Excess / surplus cash** — cash piled beyond operating needs, just sitting there. This is a **non-operating asset** — it is not helping run the business.

Your note *non-cash working capital vs working capital*: non-cash working capital is receivables + inventory − payables, *excluding* cash. We often prefer it because it isolates the operating squeeze on cash without muddling in the cash balance itself.

Practical rule (your note): if you cannot tell how much cash is operating vs excess, treat *all* of it as **surplus**. That keeps it out of the operating cash flow and values it separately.

Step 9 — How cash enters the valuation

Your note: *value of firm = value of operating + value of non-operating (cash added here)*. Exactly. The DCF of FCFF gives the value of the **operating** business only (because FCFF is *operating* cash flow). Excess cash is not in there — so you **add it back separately** at the end:

$$\text{Firm Value} = \text{PV of FCFF} \;+\; (\text{Excess cash} + \text{other non-operating assets})$$

operating value + separately-valued non-operating items

This is precisely what "value separately" meant in the Ariel and Airlead cases — surplus investments and excess cash are parked and added at the equity bridge.

Step 10 — Cash-rich companies: cost of assets vs cost of equity

Your note: *for cash-rich companies, how does cost of asset vs cost of equity vary?* Excess cash is a near-risk-free, low-return asset sitting inside the company. It **dilutes** the overall riskiness of what the firm holds. So a cash-rich company's *observed* asset return (and asset beta) looks artificially **low** — the boring cash drags the average down. Its *operating* assets might be quite risky, but the cash mask hides it.

The fix: strip the cash out first, measure the cost of capital on the *operating* business alone, then add the cash back at the end (Step 9). Otherwise you under-estimate true operating risk and over-value the business. This is *why* we bother separating cash at all.

PART 3

Cross-Border Capital Budgeting

This is the international layer — the Groupe Ariel situation, where a French parent evaluates a Mexican project. Your notes list six big questions plus the Fisher effect. Here they are, structured.

The six questions when cash flows cross a border

1. Which tax rate — domestic or foreign?

The project's cash flows are taxed where they are *earned*. Ariel's Mexican savings are taxed at Mexico's 33.33%, not France's 35% — which is exactly why using the French rate is flagged as a trap. The foreign (local) rate applies to the operating cash flows.

2. Which tax system? (How Apple pays tax in India vs the US)

Countries use different systems for taxing foreign income:

- **Worldwide / credit system** — the home country taxes global income but gives a *credit* for foreign tax paid. If the home rate exceeds the foreign rate, the company tops up the difference — effectively the **higher of the two rates** (your "we take higher rate?" note).
- **Territorial / exemption system** — the home country *exempts* foreign income entirely; you only pay the foreign rate. Your note "India follows exemption system" points here: once taxed abroad, the foreign rate is what sticks.

3. Tax holidays, sparing vs non-sparing

A **tax holiday** is when a host country waives tax to attract investment. The question is whether the home country *honours* that waiver:

- **Tax sparing** — home country pretends the foreign tax *was* paid and still gives the credit, so the company keeps the holiday's benefit.
- **Tax non-sparing** — home country ignores the holiday and taxes the income anyway; the benefit evaporates when profits come home.

4. Which cash flow — generated or repatriated?

Cash the foreign subsidiary *generates* is not always what the parent can *bring home* (repatriate). Repatriation can trigger extra taxes or hit capital controls. From the *parent's* view, what matters is the cash it can actually receive — but you often value both and discuss the gap.

5. Which currency — and Method A vs Method B?

This is the core Ariel question. There are two routes, and they **must agree**:

	Method A (home-currency route)	Method B (foreign-currency route)
Cash flows in	Foreign currency (pesos)	Converted to home currency each year
Discount rate	Foreign-currency hurdle (Fisher-adjusted)	Home hurdle (e.g. 8% euro)
FX handled by	Convert final value at spot	Forecast FX path via PPP, year by year

Which to pick when the exchange rate is unstable? Method B makes the currency path *explicit* year-by-year, so it is better when you want to stress-test currency assumptions (like Ariel's "analysts expect faster peso depreciation" sensitivity). Method A hides the currency path inside a single adjusted rate.

Building a country-risk-adjusted discount rate. Your notes give two approaches:

$$\text{Approach 1:}\quad R = Rf + \beta(\text{equity}) \times \text{US premium} + \text{Country Risk Premium}$$

a flat country premium added to everyone

$$\text{Approach 2:}\quad R = Rf + \beta(\text{equity}) \times \text{US premium} + \lambda \times \text{Country Risk Premium}$$

λ (lambda) scales the premium to THIS firm's exposure

λ (lambda) measures how *exposed this specific company* is to the country's risk. A local retailer has high λ; an exporter selling globally has low λ. Lambda is the more refined approach — it does not punish every firm equally.

6. Converting a bond spread into an equity spread

Your note: *sovereign bond spread = bond price – CAPM, or from bond rating; then spread × σ(equity)/σ(bond).* Here is what this does. A country's **sovereign bond spread** is the extra yield its government bonds pay over a safe benchmark (like US Treasuries) — the *debt-market* price of that country's risk. But equities are *more* volatile than bonds, so you scale the bond spread up to an *equity* risk premium:

$$\text{Country Equity Risk Premium} = \text{Bond Spread} \times \left[\ \sigma(\text{equity}) \ / \ \sigma(\text{bond})\ \right]$$

the volatility ratio grosses up the debt spread because stocks swing more than bonds

Can it be used elsewhere? Yes — the same "scale one market's risk to another's by their volatility ratio" logic appears whenever you translate risk between asset classes. Damodaran's website publishes these country risk premiums ready-made (your "Damodaran website" note).

The Fisher Effect — three markets, one truth

Your note: *capital market → interest rate; currency market → exchange rate; goods market → inflation rate; all interconnected.* This is the theoretical spine of the whole cross-border case. The claim is striking: **three different markets are all secretly pricing the same thing — the gap between two countries.**

If Mexico has higher inflation than France, that *single fact* shows up in all three markets at once:

- **Goods market** — Mexican prices rise 7% vs French 3% → inflation differential ≈ 4%.
- **Capital market** — Mexican interest rates run ~4% higher (lenders demand compensation for inflation eroding their money).
- **Currency market** — the peso weakens ~4% per year against the euro (so the same basket costs the same in either currency over time).

A worked example

Say you hold ₹100-equivalent of pesos today. Mexico offers 12% interest, France offers 8%. You could:

4. **Route 1 —** keep pesos, earn 12% → 112 pesos next year.
5. **Route 2 —** convert to euros today, earn 8% → 108 euros, then convert back next year.

For no free lunch to exist, these must end up equal. That forces the peso to weaken by exactly the interest differential: 112 ÷ 108 − 1 ≈ **3.7%**. The currency market *must* move to offset the interest-rate gap — otherwise everyone would pile into the higher-rate currency risk-free. That is your formula:

$$1 + \text{conversion factor} = (1 + \text{foreign interest rate}) \ / \ (1 + \text{domestic interest rate})$$

the FX shift is pinned down by the interest differential — same differential, three markets, one number

Converting the discount rate — and your two "why" questions

Your note gives the rate conversion:

$$1 + \text{WACC(foreign)} = (1 + \text{WACC(domestic)}) \times (1 + \text{conversion factor})$$

then subtract 1 to get the clean foreign-currency rate

Why the "+1" everywhere?

Not continuous compounding (a reasonable guess, but no). The +1 is because **you are compounding growth factors, not adding rates.** A rate of 8% means "multiply by 1.08." To combine two effects — the domestic return *and* the currency shift — you *multiply* their factors: (1.08) × (1.04), not 8% + 4%. The +1 turns a *rate* into a *multiplier* so they compound correctly; you subtract 1 at the end to return to a clean rate. It is the same reason the Ariel peso hurdle was (1.08)(1.07)/(1.03) − 1, not 8% + 7% − 3%. Small difference, but multiplying is the correct way to stack growth rates.

Why give inflation preference over interest? (Real interest rate logic)

The **real interest rate** is the true return after stripping out inflation: real ≈ nominal − inflation. The insight is that across countries, **real rates tend to converge** — capital flows to wherever real returns are highest until they equalize. So the *difference* in nominal interest rates between two countries is driven almost entirely by the *difference in their inflation*; the real parts cancel out. That is why, when forecasting exchange rates, we lean on the **inflation differential** (PPP) rather than the interest differential: inflation is the more *fundamental*, real-economy driver, while nominal interest is just inflation plus a roughly-common real rate. **Inflation is the signal; the rest is noise that cancels.**

QUICK REFERENCE

One-Page Formula Sheet

Everything above, compressed. Use this to revise the night before.

Part 1 — Beta

$$\beta(\text{asset}) = [D/(D+E)]\cdot\beta(\text{debt}) + [E/(D+E)]\cdot\beta(\text{equity}) \qquad [\text{general}]$$

$$\text{Unlever:}\ \ \beta(\text{asset}) = \beta(\text{equity}) \ / \ [1 + (1-t)\cdot D/E]$$

$$\text{Relever:}\ \ \beta(\text{equity}) = \beta(\text{asset}) \cdot [1 + (1-t)\cdot D/E]$$

CAPM sets β(debt) = 0 (IG debt ≈ risk-free). Use **net debt** (= debt − cash) in D/E. CAPM → higher Ke → lower, conservative value.

Part 2 — FCFF

$$\text{FCFF} = \text{EBIT}(1-t) + \text{Depreciation} - \text{CapEx} - \Delta\text{NWC} \qquad [\text{income-statement route}]$$

$$\text{FCFF} = \text{NOPAT} - \Delta\text{NWC} - \Delta\text{NFA} - \Delta\text{NOA} \qquad [\text{balance-sheet route}]$$

$$\text{Firm Value} = \text{PV of FCFF} + \text{Excess cash} + \text{Non-operating assets}$$

Cash = Profit − ΔA/R (accrual → cash). NFA = Gross Book Value − Accumulated Depreciation. If unsure, treat all cash as surplus and strip it before measuring cost of capital.

Part 3 — Cross-border

$$1 + \text{conversion factor} = (1 + \text{i_foreign})/(1 + \text{i_domestic})$$

$$1 + \text{WACC(foreign)} = (1 + \text{WACC(domestic)})\cdot(1 + \text{conversion factor})$$

$$\text{Country Equity Risk Premium} = \text{Bond Spread} \cdot \sigma(\text{equity})/\sigma(\text{bond})$$

$$\text{Country-risk discount rate} = \text{Rf} + \beta\cdot\text{US premium} + \lambda\cdot\text{Country Risk Premium}$$

Tax the cash where it is earned (foreign rate). Method B = explicit FX path (PPP), best when FX is unstable. Fisher: inflation, interest and FX differentials are three views of one country gap; lean on inflation (real rates converge).

Final word

You told me you are bad at this. You are not. These notes captured two dense lectures accurately — you were missing the connective tissue, not the ideas. Read this guide twice, work one example by hand, and it will lock in. You have got this.